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High-Temperature Fischer−Tropsch Synthesis of Light Olefins over Nano-Fe_3O_4@MnO_2 Core−Shell Catalysts

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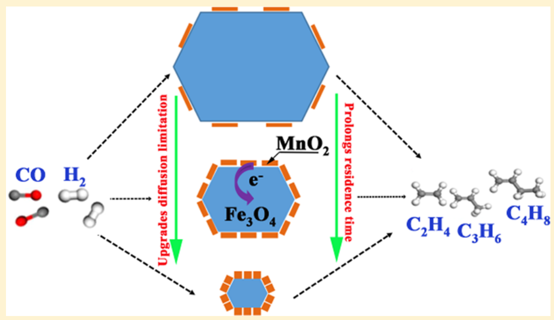
Supporting Information

ABSTRACT: A one-pot hydrothermal process was adopted to synthesize nano-Fe_3O_4@MnO_2 core−shell catalysts with diverse thicknesses of MnO_2 for the high-temperature Fischer−Tropsch (HTFT) synthesis of light olefins. Herein, the effects of the Mn amount on the morphology and catalytic performance of the catalysts were evaluated by Ar adsorption−desorption, X-ray diffraction (XRD), H_2 temperature-programmed reduction (H_2-TPR), H_2 temperature-programmed desorption (H_2-TPD), CO temperature-programmed desorption (CO-TPD), X-ray photoelectron spectroscopy (XPS), scanning electron microscopy (SEM), high-resolution transmission electron microscopy (HRTEM) equipped with energy-dispersive spectroscopy (EDS)-lining, aberration-corrected high-angle annular dark-field scanning transmission electron microscopy (HAADF-STEM) combined with EDS-mapping, and Mössbauer spectroscopy. Both EDS-lining and EDS-mapping had verified the core−shell structure in the fresh samples. The core−shell structure benefited the reduction of Fe_3O_4, electron donation, and the CO dissociative adsorption while suppressing the hydrogenation reaction. The porous MnO_2 shell inhibited the further growth and aggregation of the Fe species. Fe_3O_4@$0.3MnO_2$ showed the highest light olefin selectivity of 37.4% at a CO conversion of 91.8%. These results constitute a facile method for preparing Fe-based nanocatalysts applied in the HTFT synthesis.



1. INTRODUCTION

Light olefins ($C_2^=-C_4^=$) are regarded as important chemical raw materials and are widely used as intermediates for the production of plastics, fibers, detergents, etc.,[1,2] Conventionally, $C_2^=-C_4^=$ production is mainly done by the steam cracking of naphtha and the catalytic cracking of heavy feedstocks. Actually, light olefins are produced as byproducts in these production processes, in which the selectivity of specific olefins is relatively low.[3] Considering the focus on resources and the environment, the routes that start with syngas, which is derived from the gasification of coal and biomass or even from methane reforming, for the production of $C_2^=-C_4^=$ are of interest. Generally, syngas-based routes contain indirect processes and a direct process. Indirect processes include methanol conversion to olefins and propylene (MTO and MTP) and dimethyl ether conversion to olefins (DMTO). A direct process is achieved by Fischer−Tropsch (FT) synthesis, which is distinctly more attractive in terms of the environment and profitability to the economy.[4,5] Compared with low-temperature Fischer−Tropsch synthesis, high-temperature Fischer−Tropsch (HTFT) synthesis over Fe-based catalysts can shift the product distribution toward linear low-molecular-mass olefins.[6] Several advantages, such as low cost with abundant reserves, high selectivity for light olefins, and poisoning resistance, have made Fe-based catalysts widely used in HTFT synthesis. Nevertheless, HTFT synthesis has been limited by a low product selectivity, catalyst agglomeration, and sintering in industrial applications.[6,7] In addition, the poor mechanical stability of bulk Fe-based catalysts is another challenge of HTFT synthesis. Therefore, several researchers have employed supports to ameliorate the dispersibility of iron oxide and the mechanical degradation.[8,9] However, a cumbersome activation accompanied the supported Fe-based catalysts. In other words, highly dispersed iron species have a strong interaction with the support, retarding the reduction and carbonization of the catalysts.[1] To address this issue, researchers then turned their attention to inert supports, such as carbon nanofibers[10,11] and carbon nanotubes,[12−16] that weakly interacted with iron oxide.

The rapid development of nanomaterial science has brought about the possibility of controlling the morphology and properties of catalytic materials at the nanoscale. Recently, a superb strategy has been adopted to design core−shell

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nanostructured catalysts (encapsulated structures) to inhibit aggregation, carbon deposition, and active site blocking.[17,18] Porous silica, zeolites, and carbon materials are often used as coating materials.[19] Qiu et al.[20] used SAPO-34 zeolite to encapsulate Fe_3C nanoparticles via grafting to surface Si−OH groups, and the prepared Fe/C@SAPO was applied to the FT synthesis. The reaction results showed a high selectivity for light hydrocarbons because of the confinement effect by the SAPO-34 zeolite, which can benefit the diffusion of light hydrocarbons while hindering the formation of C_6^+ hydrocarbons. However, the pores of the silicon material were easily subject to collapse in a basic environment, hindering the syngas contact with Fe species.[19] Hence, a carbon-based coating is more attractive than previously reported materials.[21,22] Carbon materials with various functional groups (−COOH, −OH) can not only protect the Fe nanocore from harsh application situations but also improve the dispersibility of Fe species and hinder the aggregation of Fe nanoparticles.[19] Lu et al.[23] reported a facile thermal-treatment method to encapsulate Fe nanoparticles using carbon black. The FT reaction performance displayed a high liquid hydrocarbon selectivity of 65%, of which 44.8% was olefins. Later, Zhu et al.[24] synthesized Fe-Mn@C core−shell catalysts by a two-step method, which involved the encapsulation of Fe-Mn nanoparticles with oleic acid and the pyrolysis of the surface oleic acid ligands. Compared with the exposed Fe-Mn catalysts, the Fe-Mn@C catalysts showed superior stability, and their light olefin selectivity in the FT synthesis could reach 40.6% at a CO conversion of 47.5%. The authors attributed these significant improvements in catalytic performance to the confinement effect by the carbon. They hold that the thin carbon layer of the Fe-Mn@C catalyst could inhibit the reabsorption of $C_2^=$−$C_4^=$, inhibiting the chain growth. Additionally, the shell of carbon could partition the small Fe-Mn nanoparticles, thereby preventing the sintering of the catalyst. Therefore, the Fe-Mn@C catalyst possessed smaller χ-Fe_5C_2 particles, enhancing the catalytic activity and stability. These researchers have manifested that the confinement effect by shells does improve the catalytic activity and stability.

Although the carbon shell has shown outstanding performances in FT synthesis, it is limited by loss at high temperatures and by the difficulty involved in removing carbon deposition when the catalyst is recycled for reuse.[19] Manganese is an efficient promoter in FT synthesis and has been extensively investigated for many years.[25−31] Some research has indicated that Mn can not only promote the dispersion of Fe but also prevent Fe from deactivation through carbon deposition.[32,33] In addition, it is widely acknowledged that Mn can improve the $C_2^=$−$C_4^=$ selectivity while suppressing the formation of methane.[29] However, Mn is inclined to interact with iron species to form a solid solution phase, which may restrain the reduction and carbonization of the catalysts, decreasing the FT synthesis activity.[27,34] Interestingly, Zhang et al.[35] prepared core−shell Fe_2O_3@MnO_2 spindles by a two-step hydrothermal method. This core−shell design can not only weaken the interaction between Mn and Fe species but also take advantage of the promotion of Mn toward olefin selectivity. The core−shell catalysts showed a high selectivity toward C_5^+ hydrocarbons while maintaining a low selectivity toward methane in the FT synthesis. Furthermore, Zhao et al.[36] proposed a one-pot process to fabricate tunable Fe_3O_4@MnO_2 nanoplates for water purification. Recently, Wang et al.[37] employed this novel core−shell Fe_3O_4@MnO_2 in the FT synthesis of linear α-olefins. This catalyst exhibited 75.5% CO conversion and a high selectivity of 65.0% for C_4^+ alkenes, 91.0% of which were α-olefins. However, the reaction occurred at a low temperature and a low gas hourly space velocity (GHSV), which means that the yield of olefins was low, although selectivity was high. Furthermore, the product distribution was wide, increasing the difficulty of product separation.

Not long ago, the "confinement effect" of carbon materials encapsulating nano-Fe particles was proposed, which inspired the design of core−shell catalysts.[38,39] Encapsulated metal nanoparticles exhibit a small particle size. In addition, the pore size and thickness of the shell exhibit great effects on the diffusion behaviors of reactants and products. As far as we know, the structural effect of the core−shell configuration with a mesoporous MnO_2 shell in the HTFT synthesis of light olefins has not been reported. In this study, core−shell nano-Fe_3O_4@MnO_2 catalysts, with various thicknesses of the MnO_2 layer, were prepared by a facile one-pot hydrothermal process, which simplifies the two-step hydrothermal preparation method of Zhang et al.[40] The effects of the Mn content on the structure, morphology, and catalytic performance of the nano-Fe_3O_4@MnO_2 catalysts are discussed by combining various characterization techniques including Ar adsorption−desorption, X-ray diffraction (XRD), H_2 temperature-programmed reduction (H_2-TPR), H_2 temperature-programmed desorption (H_2-TPD), CO temperature-programmed desorption (CO-TPD), X-ray photoelectron spectroscopy (XPS), scanning electron microscopy (SEM), high-resolution transmission electron microscopy (HRTEM), energy-dispersive spectroscopy (EDS)-lining, EDS-mapping, and Mössbauer spectroscopy. The activity and selectivity of the products are discussed and compared with those of the traditional coprecipitation catalyst. Furthermore, we emphasize the confinement effect of the shells with different thicknesses on the HTFT product distribution, activity, and stability. These discussions fill the gaps in the studies performed by other researchers, and a preliminary relationship of the catalyst structure and performance with the production of $C_2^=$−$C_4^=$ in HTFT synthesis is established.

2. EXPERIMENTAL SECTION

2.1. Materials. Iron(II) sulfate heptahydrate ($FeSO_4\cdot 7H_2O$, 99.95% purity), poly(vinylpyrrolidone) (PVP, K88-96), sodium hydroxide (NaOH, 98% purity), and manganese nitrate solution ($Mn(NO_3)_2$, 50% wt % in H_2O) were of guaranteed grade or analytical grade and purchased from Shanghai Macklin Biochemical Co., Ltd. Potassium permanganate ($KMnO_4$, 99.5% purity) was obtained from Titan Scientific Co., Ltd, Shanghai. Deionized water (DI water) was collected from the division of laboratory and facility management in school.

2.2. Syntheses of Catalysts. *2.2.1. Synthesis of Nano-Fe_3O_4.* The nano-Fe_3O_4 was synthesized using a one-pot hydrothermal process. First, 2.9 g of NaOH was dissolved in 200 mL of DI water under heating and stirring at 70 °C. Meanwhile, 10 g of $FeSO_4\cdot 7H_2O$ and 6.0 g of PVP were dissolved in 100 mL of DI water under magnetic stirring until a homogeneous solution was formed. Then, the mixed solution was dropped into the NaOH solution under stirring at 70 °C, generating a dark-green suspension. The suspension was aged with mother liquid for 2 h at 70 °C. After centrifugation and washing with DI water and absolute ethanol, successively, the resulting solid sample was dried at 110 °C for 12 h and heat-

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treated at 400 °C for 4 h. These procedures were carried out in a tube furnace under the flow of high-purity nitrogen (99.9995% purity, i.e., 5.5 quality). Finally, the heat-treated product was screened out for 40−60 mesh particles for the HTFT reaction. The nanocatalyst was named Fe_3O_4.

2.2.2. Synthesis of Core−Shell Nano-Fe_3O_4@MnO_2. The core−shell nano-Fe_3O_4@MnO_2 was synthesized using a one-pot hydrothermal process with some modifications.[40] After generating the above dark-green suspension, 50 mL of $KMnO_4$ solutions, containing $KMnO_4$ in various quantities (0.57, 1.1, 1.7, and 2.3 g), was dripped into the suspension while stirring slowly, producing a dark-brown precipitate. The sediment was aged with the mother liquid for 2 h at 70 °C. After centrifugation and washing with DI water and absolute ethanol several times, successively, the sample was dried and heat-treated similar to that for Fe_3O_4. To avoid oxidation of the product, the drying and heat-treatment processes were performed in a tube furnace under a high-purity N_2 (99.9995% purity, i.e., 5.5 quality) flow. Finally, the heat-treated product was screened out for 40−60 mesh particles for the HTFT reaction. The nanocatalysts with the different Fe/Mn molar ratios were named Fe_3O_4@0.1MnO_2, Fe_3O_4@0.2MnO_2, Fe_3O_4@0.3MnO_2, and Fe_3O_4@0.4MnO_2.

2.2.3. Synthesis of Nano-Fe_3O_4-0.3Mn. The nano-Fe-Mn spinel was prepared by the traditional coprecipitation method. Typically, 2.9 g of NaOH was dissolved in 200 mL of DI water under stirring and heating at 70 °C. Meanwhile, a homogeneous solution with an Fe/Mn molar ratio of 0.3 was prepared, containing 10 g of $FeSO_4·7H_2O$, 3.9 g of $Mn(NO_3)_2$ solution, 6.0 g of PVP, and 100 mL of DI water. Then, the mixed solution was dropped into the NaOH solution under stirring at 70 °C, generating a dark-brown sediment. The sediment was aged for 2 h at 70 °C. After centrifugation and washing, successively, the resulting solid sample was dried and heat-treated similar to that for Fe_3O_4, and these procedures were carried out in a tube furnace under a flow of high-purity nitrogen (99.9995% purity, i.e., 5.5 quality). Finally, the heat-treated product was screened out for 40−60 mesh particles for the HTFT reaction. The nanocatalyst was named Fe_3O_4-0.3Mn.

2.3. Characterization. The specific surface areas, average pore sizes, and total pore volumes of the samples, as well as the adsorption isotherm type, were surveyed by Ar physisorption at −186 °C on a Micrometrics ASAP 2020. The samples' specific surface areas were calculated by the Brunauer−Emmett−Teller (BET) method. The average pore sizes and total pore volumes were measured by the Barrett−Joyner−Halenda (BJH) approach, based on the Kelvin equation.

X-ray diffraction (XRD) patterns were acquired on Rigaku D/MAX2550 VB/PC XRD equipped with Cu Kα radiation (λ = 0.154 nm) at 18 kW and 450 mA.

The contents of Mn and Fe in the samples were quantified by inductively coupled plasma atomic emission spectroscopy (ICP-AES), which was measured on Agilent 725ES.

The H_2 temperature-programmed reduction (H_2-TPR), H_2 temperature-programmed desorption (H_2-TPD), and CO temperature-programmed desorption (CO-TPD) tests were conducted on a Micrometrics AutoChem II 2920 chemisorption apparatus.

Scanning electron microscopy (SEM) was investigated with FEI Inspect F50. High-resolution transmission electron microscopy (HRTEM) micrographs and energy-dispersive spectroscopy lining (EDS-lining) measurements were obtained on a JEM-2100F microscope armed with an X-ray EDS detector. The high-angle annular dark-field scanning transmission electron microscopy (HAADF-STEM) measurement was carried out on a double-corrected FEI Titan Themis scanning/transmission electron microscope operated at 300 keV and equipped with a probe spherical aberration corrector. Energy-dispersive spectroscopy mapping (EDS-mapping) was carried out on a FEI Super X-EDS system. The particle size distribution histograms of the samples were obtained by randomly measuring at least 100 iron particles from the HRTEM images.

X-ray photoelectron spectroscopy (XPS) spectra of the fresh and reduced catalysts were obtained on a Thermo ESCSLAB 250 X-ray photoelectron spectrometer using Al Kα (1486.8 eV, 150 W, 500 μm) as the excitation source.

The Mössbauer spectra of the samples after the reactions were acquired on a Mössbauer spectrometer (Wissel, Germany) equipped with 57Co in a Pd matrix at room temperature. The least-squares method was used to fit the spectra.

2.4. Catalyst Testing and Product Analysis. The HTFT reaction was conducted in a 500 mm length and 10 mm i.d. fixed-bed reactor. Typically, 0.5 g of catalyst was mixed with 1.0 g of quartz sand of 40−60 mesh particles. Before the HTFT synthesis, the sample was reduced by a 50%H_2/50%N_2 (5000 mL/(h·g_{Cat}), 0.1 MPa, 350 °C) flow for 10 h. Afterward, the reactor was cooled to 200 °C by high-purity N_2. Subsequently, syngas (60%H_2/30%CO/10%N_2) was introduced to the reactor. The FT reaction conditions were as follows: a gas hourly space velocity (GHSV) of 11000 mL/(h·g_{Cat}), a temperature of 340 °C, and a pressure of 1.5 MPa.

The tail gas was analyzed inline by gas chromatography (GC) Agilent 7890B. After being separated on HayeSep Q and 5A molecular sieve packed columns, N_2, CO_2, CO, and H_2 were detected by two thermal conductivity detectors (TCDs). The C_1−C_6 hydrocarbons in the tail gas were separated on the HP-AL/S capillary column and were detected by flame ionization detector (FID). Meanwhile, the aqueous, oil, and wax phases on the reaction products were collected after the reaction stabilized for 48 h and then analyzed offline by GC Agilent 7890A equipped with an Agilent 5975C mass spectrometer.

CO conversion (X_{CO},%), CO_2 selectivity (S_{CO_2},%), and hydrocarbon distribution ($S_{C_iH_j}$,%) of each hydrocarbon were calculated by formulas 1−3, respectively

$$X_{CO}(\%) = \frac{N_{in,CO} - N_{out,CO}}{N_{in,CO}} \times 100 \tag{1}$$

$$S_{CO_2}(\%) = \frac{N_{out,CO_2}}{N_{in,CO} - N_{out,CO}} \times 100 \tag{2}$$

$$S_{C_iH_j}(\%) = \frac{iN_{out,C_iH_j}}{\sum_{i=1}^{n} iN_{out,C_iH_j}} \times 100 \tag{3}$$

where $S_{C_iH_j}$ is based on a carbon atom basis and N_{in} and N_{out} represent molar flow rate of the inlet and outlet gas, respectively.

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3. RESULTS AND DISCUSSION

3.1. Catalyst Characterization. *3.1.1. Structure and Composition of the Catalysts.* The variations in the phase structures of the as-prepared samples after heat treatment were further studied by XRD. Figure 1A indicates that the five samples can be assigned to the face-centered cubic crystal of Fe_3O_4, which is consistent with the standard JCPDS card (No. 19-0629). The diffraction peaks at 2θ of 30.1, 35.4, 43.1, 53.4, 56.9, and 62.5° can be assigned to the (220), (311), (400), (422), (511), and (440) planes of the Fe_3O_4 phase, respectively. With the gradual increase in the content of Mn, the intensity of the Fe_3O_4 diffraction peaks significantly decreases. Meanwhile, new diffraction peaks located at 2θ of 33.1, 40.8, and 49.5° appear and can be attributed to MnO_2. This result indicates that Mn is in the form of MnO_2, rather than dissolved into the Fe_3O_4 phase to form Fe-Mn spinel phases. These results reveal that Fe_3O_4 is well dispersed by the addition of Mn, which is beneficial for the stability of the small crystals.[41] Therefore, the average crystalline size of Fe_3O_4 gradually decreases, as shown in Table 1.

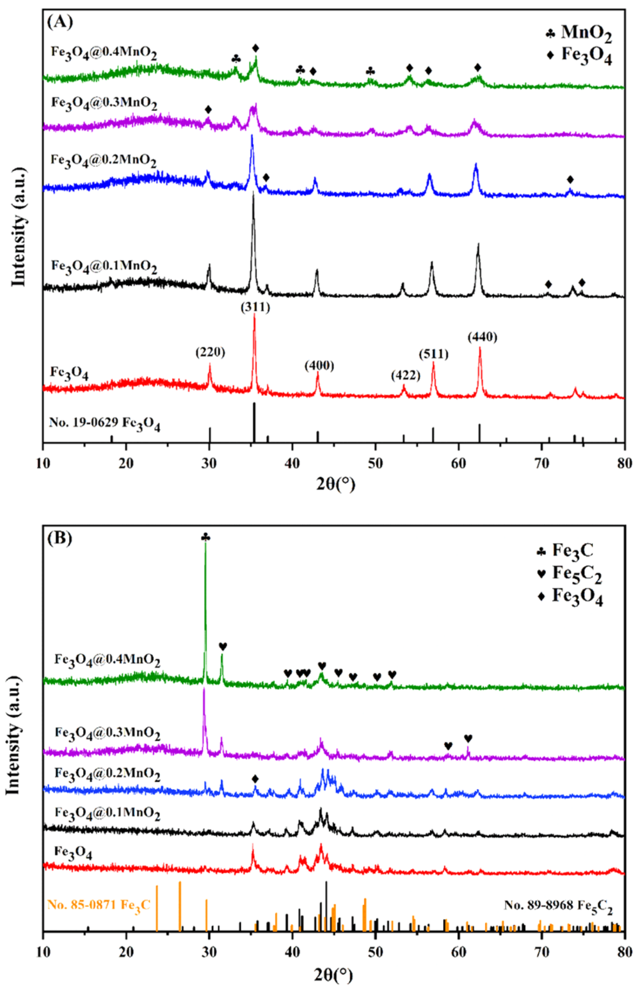

Figure 1. XRD patterns of the samples: (A) after heat treatment and (B) after reaction.

In addition, the XRD pattern of Fe_3O_4-0.3Mn prepared by the traditional coprecipitation method is presented in Figure S1. Noticeably, there is no diffraction peak for MnO_2 in the XRD pattern of Fe_3O_4-0.3Mn. The XRD diffraction peak intensity of Fe_3O_4-0.3Mn is significantly stronger than that of Fe_3O_4@0.3MnO_2. These results can be attributed to the Mn atoms being dissolved in the Fe_3O_4 phase, forming an Fe-Mn spinel phase. The strong interaction between Fe and Mn can stabilize the Fe_3O_4, enhancing its crystallinity. In contrast, the crystallinity of Fe_3O_4@0.3MnO_2 is relatively low, which may be due to the absence of strong interactions between the Fe and Mn of the core−shell structure. Additionally, the detection of Fe_3O_4 can be inhibited because of the coverage of the MnO_2 layer on the surface of catalysts.

Table 1. Component and Textural Properties of the Fresh Catalysts

catalyst	Mn/Fe molar ratio[a] (%)	S_{BET} ($m^2 \cdot g^{-1}$)	V_p ($cm^3 \cdot g^{-1}$)	D_p (nm)	$d_{Fe_3O_4}$[b] (nm)	particle size[c] (nm)
Fe_3O_4	0	20.4	0.13	25.4	38.7	42.8
Fe_3O_4@0.1MnO_2	10.1	36.6	0.19	21.8	30.0	32.1
Fe_3O_4@0.2MnO_2	20.3	44.2	0.22	20.9	25.5	26.2
Fe_3O_4@0.3MnO_2	29.6	57.0	0.30	18.5	21.9	22.3
Fe_3O_4@0.4MnO_2	40.2	57.6	0.34	19.6	15.9	19.0
Fe_3O_4-0.3Mn	29.7	24.6	0.15	22.8	29.9	33.4

[a]Surveyed by ICP-AES. [b]Calculated by Scherrer's equation. [c]Determined by HRTEM.

Figure 1B displays the XRD diffraction peaks of the samples after reacting with syngas for 48 h. The XRD spectra indicate that Fe_3O_4 and χ-Fe_5C_2 (No. 89-8968) are the main phases of Fe in the spent Fe_3O_4 catalyst. Distinctly, the Fe_3O_4 peak located at 2θ of 35.42° gradually weakens until it disappears with the increase in the content of Mn. Conversely, the intensity and number of the χ-Fe_5C_2 peaks increase significantly. This result indicates that Mn promotes the transformation of Fe_3O_4 to χ-Fe_5C_2 in the HTFT synthesis. Notably, a novel diffraction peak located at 2θ of 29.66° appears as the content of Mn increases to 0.2, which can be attributed to θ-Fe_3C (No. 85-0871). With further increases in the content of Mn, the intensity of the θ-Fe_3C peak significantly increases, while the intensity of the χ-Fe_5C_2 peak decreases. This change may be due to the excellent dispersity of Fe species and the slight decrease in the χ-Fe_5C_2 content at higher Mn contents. Further information regarding the iron species will be gained in the following discussion by Mössbauer spectroscopy.

The molar ratios of Mn to Fe in the catalysts detected by ICP-AES are also displayed in Table 1 and indicate that the contents of Mn and Fe are close to the nominal values of the samples during preparation.

Low-temperature Ar physisorption was used to survey the textural characteristics of the samples. The Ar adsorption−desorption isotherms (Figure 2) display typical characteristics of type IV isotherms accompanied by an evident type H1 hysteresis loop, belonging to a mesoporous structure.[42] The BET surface areas, total pore volumes, and average pore sizes of the fresh samples are displayed in Table 1. Clearly, the BET surface area increases from 20.4 to 57.6 $m^2 \cdot g^{-1}$ as the content of Mn increases from 0 to 0.4. Meanwhile, the pore volume increases from 0.13 to 0.34 $cm^3 \cdot g^{-1}$, while the average pore size shows a gradual decreasing trend. These data are consistent with the XRD results, further confirming that the addition of

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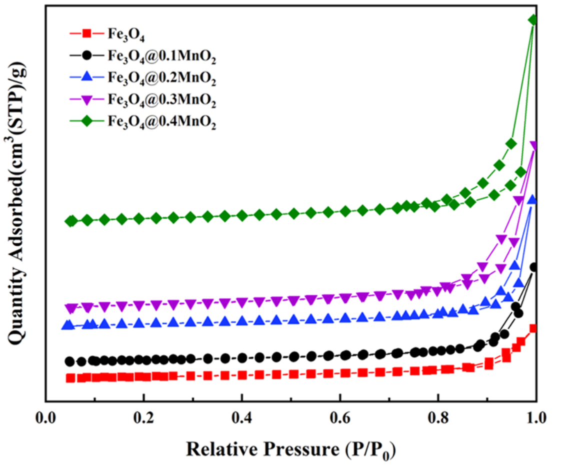

Figure 2. Ar adsorption−desorption isotherms of the fresh samples.

Mn benefits the dispersity of Fe_3O_4 and reduces the grain size of Fe_3O_4. In addition, the BET surface area of Fe_3O_4-0.3Mn prepared by the traditional coprecipitation method is approximately half of that of Fe_3O_4@$0.3MnO_2$, which may be due to the abundant pores of the surface MnO_2 layer.

3.1.2. Morphology and Elemental Distribution. To further inspect the morphological and structural information of the samples, SEM, HRTEM, HAADF-STEM, STEM EDS-lining, and EDS-mapping measurements were adopted to survey the catalysts. Figure S2 exhibits the SEM images of five samples. Only spherical nanoparticles can be observed for the as-prepared single Fe_3O_4 catalyst, which is attributed to the face-centered cubic Fe_3O_4. Interestingly, some nanoplates appear as a consequence of the Mn addition and are marked by red dotted ellipses in Figure S2B. In addition, more nanoplates appear with the increasing contents of Mn species, manifesting that the Mn species facilitate the formation of nanoplates.

HRTEM images were also used to illustrate the morphological changes and are depicted in Figure 3A−F, and the insets are particle size distribution histograms of samples. Table 1 lists the mean particle sizes of the six catalysts. Figure 3 illustrates that the Fe_3O_4 samples without the MnO_2 shell are prone to agglomeration and form large agglomerates. However, the samples become smaller and well dispersed with the introduction of the MnO_2 layer. The MnO_2 shell may restrict the growth and aggregation of Fe_3O_4 through space confinement. In particular, some nanoplates can be observed with the addition of Mn. The number of nanoplates has significantly increased as the molar ratio of Mn increases to 0.3. Interesting, Fe_3O_4-0.3Mn only shows a spherical shape. This result indicates that the core−shell structure promotes the transition of spherical morphology to plates.

In addition, the typical Fe_3O_4@$0.3MnO_2$ was further investigated by the smaller scale in Figure 4. As measured from the image, the lattice spacings of 2.53 and 4.84 Å belong to the (311) and (111) planes of Fe_3O_4, respectively. Furthermore, STEM EDS-lining analysis was adopted to inspect the distributions of Fe and Mn in the catalysts. Figure 4B,C shows the EDS-lining measurement of Fe_3O_4@$0.3MnO_2$, which can show the elemental distributions in the catalyst and can approximately measure the thicknesses of the inner and outer layers.[43] When the Mn signal begins to increase, the characterized spot moves on to the edge of the nanoplate. Figure 4C presents that the Mn signal is gradually enhanced, reaching a maximum at the edge of the nanoplate. Then, the Mn signal weakens and reaches its lowest level. These results

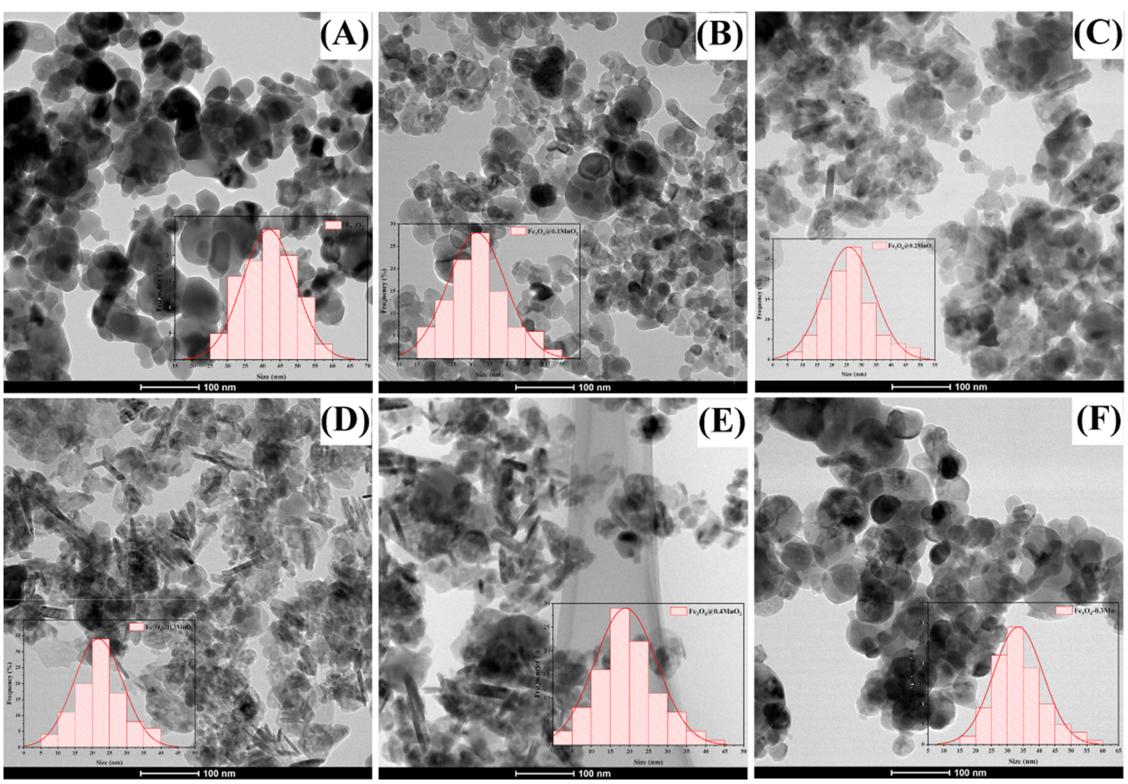

Figure 3. HRTEM images and particle size distribution histograms of synthesized samples (A) Fe_3O_4, (B) Fe_3O_4@$0.1MnO_2$, (C) Fe_3O_4@$0.2MnO_2$, (D) Fe_3O_4@$0.3MnO_2$, (E) Fe_3O_4@$0.4MnO_2$, and (F) Fe_3O_4-0.3Mn.

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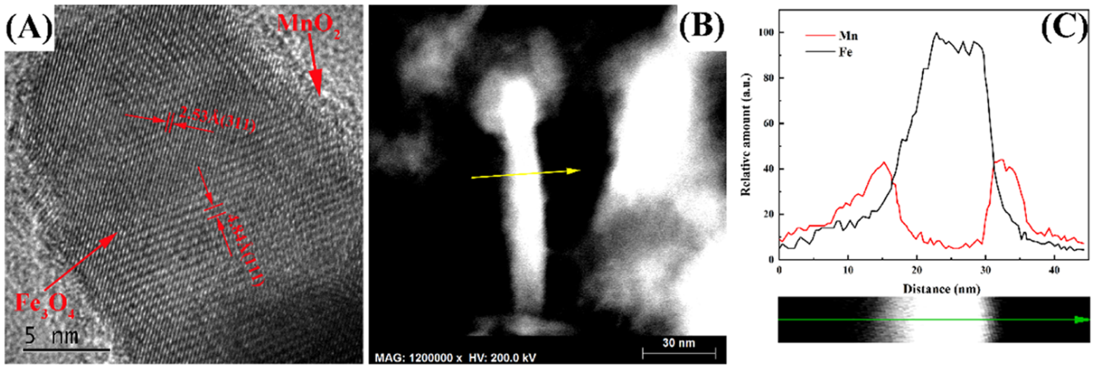

Figure 4. (A) HRTEM image of as-synthesized $Fe_3O_4@0.3MnO_2$; (B) and (C) STEM EDS-lining images of $Fe_3O_4@0.3MnO_2$.

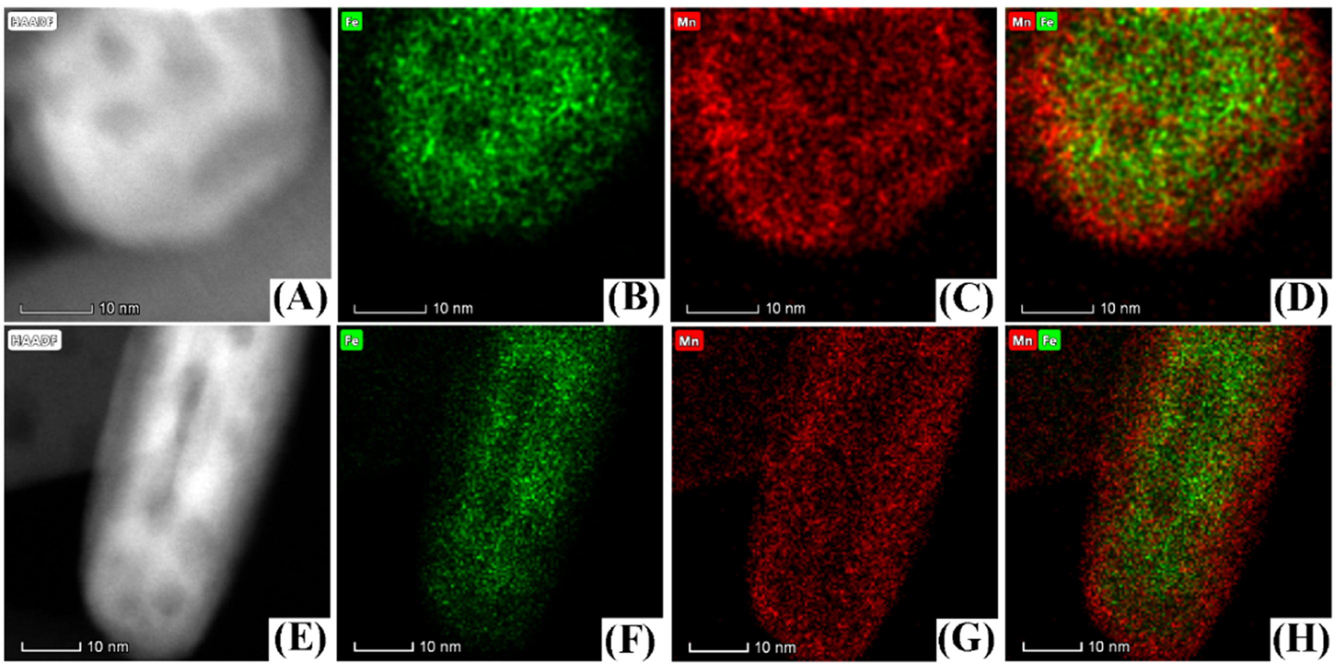

Figure 5. Aberration-corrected HAADF-STEM and corresponding EDS-mapping images (A–D) $Fe_3O_4@0.1MnO_2$ and (E–H) $Fe_3O_4@0.3MnO_2$; Fe (green) and Mn (red).

indicate Mn enrichment of the external surface layer of the nanoplate. However, the Fe signal slowly increases first and then sharply enhances until it reaches the maximum, after which it maintains this amount. Subsequently, the Fe signal sharply weakens. These variations show that Fe is mainly located in the inner layer of the nanoplate. Noticeably, the Mn signal is gradually enhanced as the Fe signal sharply weakens. This outcome means that the characterized spot moves out of the core of the nanoplate and onto the shell of the nanoplate. Conversely, the gradual decrease in the Mn signal reveals that the characterized spot moves away from the edge of the nanoplate. Overall, the EDS-lining further verifies the adherence of MnO_2 to the surface of Fe_3O_4, forming a distinct core–shell structure.

To further verify the existence of the core–shell structure, $Fe_3O_4@0.1MnO_2$ and $Fe_3O_4@0.3MnO_2$ were characterized by aberration-corrected HAADF-STEM. Figure 5 shows the HAADF-STEM and corresponding EDS-mapping images. The result indicates that Mn is enriched on the samples' surface, while Fe is rich inside the samples, further verifying the existence of the core–shell structure. According to Figure 5D,H, we can measure the thickness of the MnO_2 shells of $Fe_3O_4@0.1MnO_2$ and $Fe_3O_4@0.3MnO_2$ as 1.2 and 3.1 nm, respectively.

Figure 6A–D clearly illustrates $Fe_3O_4@MnO_2$ catalysts with various MnO_2 shell thicknesses. The $Fe_3O_4@0.1MnO_2$ is spherical, and the thickness of the MnO_2 layer is 1.1 nm, which is close to the result of EDS-mapping in Figure 5. Figure 3C indicates that some obvious nanoplates appeared in the $Fe_3O_4@0.2MnO_2$, while Figure 6B indicates that these plate-shaped Fe_3O_4 cores present a thickness of 22.3 nm. Meanwhile, a rough and amorphous MnO_2 layer with a thickness of 1.9 nm is attached to the surface of the Fe_3O_4 cores, forming a distinct core–shell structure. With the further addition of Mn, the width of the nanoplate cores decreases from 22.3 to 7.7 nm, while the thickness of the nanoplate shells relatively increases. In addition, Figure 6D illustrates that the thicker MnO_2 layer makes the surface and contour of the nanoplates rougher, which indicates more holes.

In addition, the morphologies of the used samples after 48 and 288 h of reaction are shown in Figure S3. There are significant aggregation and growth of particle size in the Fe_3O_4 and Fe_3O_4-0.3Mn catalysts. In contrast, the spent $Fe_3O_4@0.3MnO_2$ catalyst remains granular. The morphology of the

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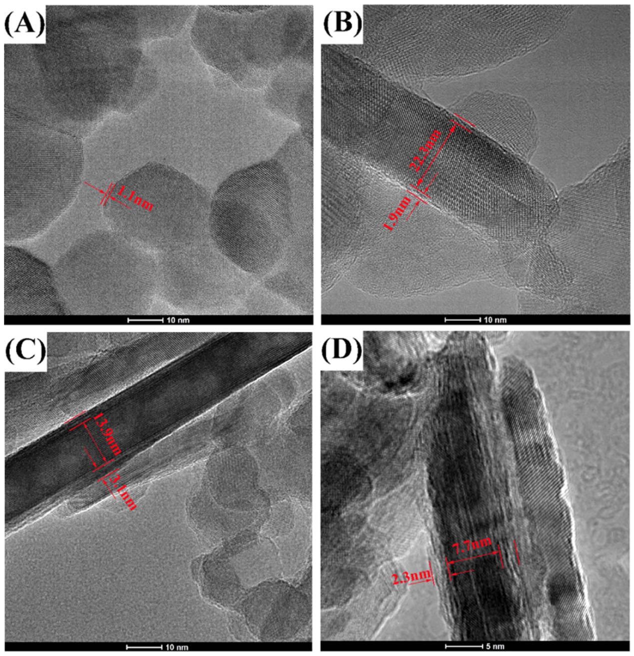

Figure 6. HRTEM image of as-synthesized catalysts (A) Fe_3O_4@$0.1MnO_2$, (B) Fe_3O_4@$0.2MnO_2$, (C) Fe_3O_4@$0.3MnO_2$, and (D) Fe_3O_4@$0.4MnO_2$.

spent catalyst after 288 h is similar to that of 48 h, indicating that the porous MnO_2 shell inhibits the further growth and aggregation of the Fe species.

3.1.3. Electronic Effect. XPS measurements were implemented to survey the surface electronic structure of the samples. Figure 7A presents two typical peaks with binding energies near 711.3 and 724.7 eV, which can be assigned to Fe $2p_{3/2}$ and Fe $2p_{1/2}$, respectively.[44] Noticeably, both Fe $2p_{3/2}$ and Fe $2p_{1/2}$ peaks shift slightly toward lower binding energies with the increase in MnO_2. In contrast, a significant increase is observed in the binding energy of Mn 2p, which is shown in Figure 7B. The different variations in the binding energies between Fe 2p and Mn 2p suggest that MnO_2 donates electrons to Fe_3O_4. In terms of the peak intensity, the Fe 2p peak presents a decreasing trend, while the Mn 2p peak shows an increasing trend, which can be attributed to the decrease in the thickness of Fe_3O_4 and to the increase in the thickness of MnO_2.[36] In addition, XPS was also adopted to survey the phase evolution on the surface layer of the reduced samples, as shown in Figure 7C. Compared with the fresh samples, both Fe $2p_{3/2}$ and Fe $2p_{1/2}$ peaks of the reduced catalysts shift toward lower binding energies. Especially, a weak peak at 706.6 eV is noted and can be assigned to the reduced metallic Fe^0, revealing that partial ionic iron has been reduced to metallic Fe.[45,46] The intensity of the Fe^0 peak exhibits an enhancement with the increase in the content of Mn, confirming the promotion of MnO_2 via the reduction of Fe_3O_4.

3.1.4. Catalyst Reducibility. Figure 8 presents the H_2-TPR profiles of the different samples. It was verified that the iron oxides can be completely reduced to α-Fe in the TPR processes when the reduction temperature increases to 700 °C.[47] However, some significant differences exist in the TPR processes of the five samples. There is a small peak and a broad overlapping peak in the H_2-TPR profile of the Mn-free sample. The small peak located at 250−300 °C can be assigned to the reduction of a small amount of Fe_2O_3, originating from the

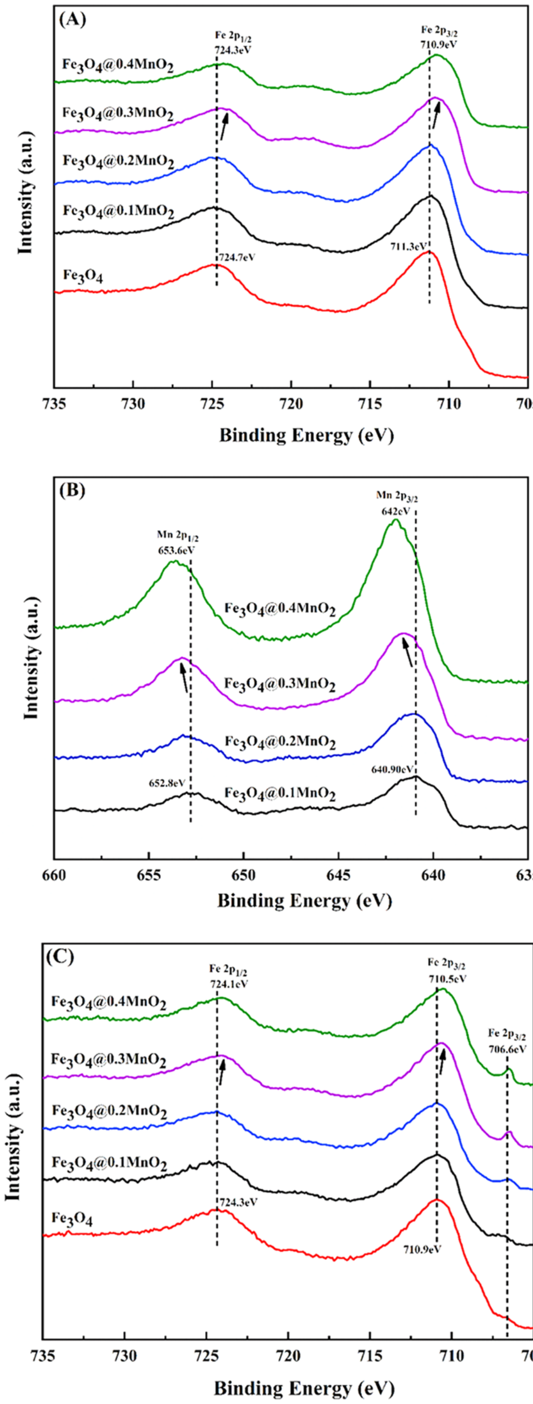

Figure 7. (A) Fe 2p XPS spectra of the fresh samples; (B) Mn 2p XPS spectra of the fresh samples; (C) Fe 2p XPS spectra of the reduced samples.

oxidation of surface Fe_3O_4 to Fe_2O_3, whose reduction temperature is considerably similar to that in the literature.[48] Note that no diffraction peak of Fe_2O_3 was detected by XRD, which may have been due to the limited amount of Fe_2O_3 in the Fe_3O_4 background being beyond the detection limit of XRD. In addition, distinct overlapping peaks centered at 500 and 600 °C can be ascribed to the continual reduction of Fe_3O_4 to FeO and FeO to α-Fe, respectively.[49,50] MnO_2 has been confirmed to reduce to MnO when the reduction temperature is approximately 400 °C.[50] Therefore, the reduction peak of the MnO_2-coated Fe_3O_4 catalysts located

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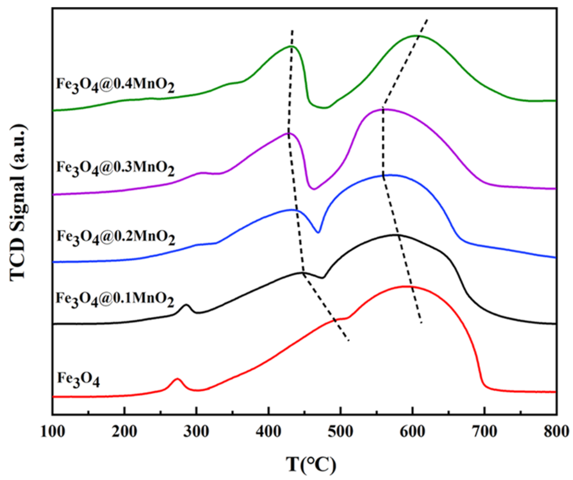

Figure 8. H_2-TPR profiles of the fresh samples.

in the 350−450 °C range can be attributed to the recombination of the reduced MnO_2 and Fe_3O_4. Notably, the overlapping peak between 350 and 700 °C tends to shift to lower temperatures with the increase in the content of Mn, certifying that the Fe_3O_4 nanoparticles coated by MnO_2 are easier to reduce than those of the bulk Fe_3O_4 sample, and the consequence may be attributed to the enhanced ability of MnO_2 donating electrons to Fe_3O_4. This result is different from that in previous studies in that the enhanced interaction between Fe and Mn caused by amalgamating Mn ions into the Fe lattice can inhibit catalyst reduction.[27,51] However, the reduction peak shifts to higher temperatures as the content of Mn increases to 0.4, and this shift is due to the excess MnO_2 covering the surface of Fe_3O_4 and blocking the contact between Fe_3O_4 and hydrogen. Furthermore, the two distinct peaks split from the overlapping peak are gradually divided with the further increase in Mn species, which can be caused by the fact that the increase in the MnO_2 content changes the reduction process in the 350−450 °C range and slightly increases the interaction between Fe and Mn.[31]

3.1.5. H_2 and CO Chemisorption. Both H_2-TPD and CO-TPD were used to investigate the chemisorption properties of the five samples. Figure 9 depicts the H_2 desorption peaks over the H_2-reduced catalysts. Previous studies have revealed that H_2 adsorption is favored in an electron-deficient metal.[52] In this regard, the H species that desorb below 130 °C are inclined to take up the superficial hollows or be located on top of weak adsorption locations, indicating a weak Fe−H bond energy, which is termed H_α. The H species that desorb in the 130−300 °C range can be attributed to the hydrogen desorbing from the deep hole or defect locations in surfaces of metal Fe, which are termed H_β.[53] Additionally, the H_2 desorption peak that exceeds 300 °C can result from the decomposition of OH^- on the surface of Fe oxides that are difficult to reduce, which is termed H_γ.[54] The Mn-free catalyst shows a broad shoulder peak centered at 600 °C, which is designated H_γ, indicating the hard reduction of Fe species and the strong chemisorption of H_2. Interestingly, this strong H_2 desorption peak above 300 °C weakens rapidly and finally disappears as the Mn content increases from 0 to 0.4. Meanwhile, two new H_2 desorption peaks, which are designated H_α and H_β, appear below 300 °C. Table 2 lists the desorbed amounts of H_2 over various Fe sites. The overlapping peaks were fitted with Gaussians, which are shown in Figure S4. Table 2 shows that the H species desorbed over the Fe_3O_4 sample is mainly H_γ species, indicating strong chemisorption of H_2. The amount of H_γ species decreases sharply with the addition of Mn. The amounts of H_α and H_β species over Fe_3O_4@MnO_2 catalysts are higher than that of the Fe_3O_4 sample, which can be attributed to the notable increase in specific surface area with the introduction of Mn. The amount of H_α species decreases from 23.2 to 11.2 μmol/g as the Mn content increases from 0.1 to 0.3. The above results fully uncover that the hydrogenation capacity of the samples gradually weakens, which is attributed to the enhanced ability of MnO_2 donating electrons to Fe_3O_4 to reduce the electron vacancies of the surface iron species.

Table 2. H_2 and CO Uptakes by Samples[a]

catalyst	H_2 uptakes (μmol/g) H_α	H_β	H_γ	CO uptakes (μmol/g) molecular CO	dissociative CO
Fe_3O_4	1.0	1.8	41.2	29.1	0
Fe_3O_4@0.1MnO_2	23.2	36.1	7.1	24.2	37.7
Fe_3O_4@0.2MnO_2	15.4	53.2	3.1	24.8	76.9
Fe_3O_4@0.3MnO_2	11.2	30.1	0	32.9	144.9
Fe_3O_4@0.4MnO_2	14.2	34.5	0	43.2	200.7

[a]H_2 and CO uptakes obtained by integration of the corresponding peak based on unit mass of sample mass.

In addition, the H_2-TPD of Fe_3O_4-0.3Mn is depicted in Figure S5. The intensities of the H_2 desorption peaks located at 100 and 200 °C for Fe_3O_4-0.3Mn are both stronger than that for Fe_3O_4@0.3MnO_2. Additionally, there is a H_2 desorption peak that exceeds 300 °C for Fe_3O_4-0.3Mn, which is not observed in the plot of Fe_3O_4@0.3MnO_2. These results further suggest that the core−shell configuration enhances the ability of MnO_2 donating electrons to Fe_3O_4, which can reduce the electron vacancies of surface iron species, suppressing the hydrogenation reaction.

The CO-TPD was used to investigate the adsorption morphologies and the variations in the amount of CO. The desorption peaks of the reduced catalysts are presented in Figure 10. Only one significant CO desorption peak is located in the 70−150 °C range for the bulk Fe_3O_4 sample, which corresponds to the molecular CO adsorbed state, indicating weak CO adsorption.[55] With the addition of MnO_2, a new CO

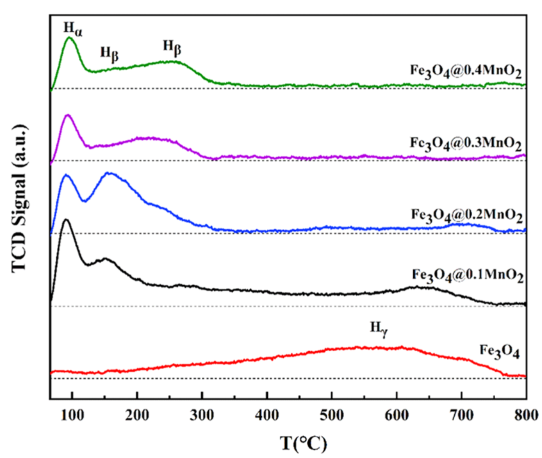

Figure 9. H_2-TPD profiles of H_2-reduced catalysts.

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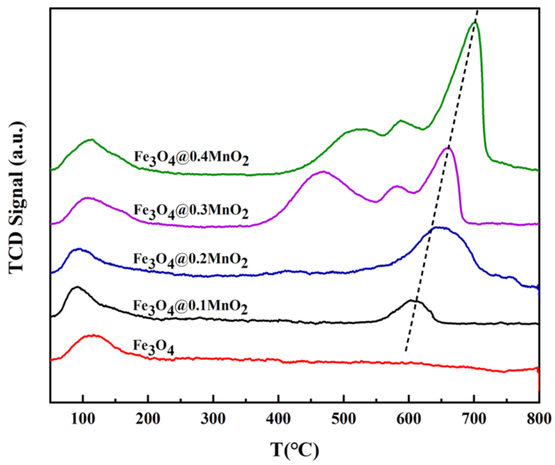

Figure 10. CO-TPD profiles of the H_2-reduced catalysts.

desorption peak, which can be ascribed to the desorption of dissociative CO and implies strong CO adsorption, appears in the 550−650 °C range.[16,56] With the gradual increase in the MnO_2 content, the peak for dissociative CO adsorption shifts to higher temperatures, and the peak intensity is enhanced, indicating that MnO_2 strengthens the adsorption of CO species at Fe sites. Furthermore, two new dissociative CO adsorption peaks located in the 400−550 and 550−600 °C ranges appear when the MnO_2 content increases to 0.3, revealing more active sites for CO dissociation and adsorption. The amounts of corresponding CO desorbed in different adsorption morphologies are also summarized in Table 2. The Fe_3O_4 sample only contains the desorption of the molecular CO. The amount of desorption of the molecular CO increases from 24.2 to 43.2 μmol/g as the MnO_2 content increases from 0.1 to 0.4. The amount of desorption of the dissociative CO displays a monotonic increase tendency with the gradual increase in the MnO_2 content. These results can be attributed to the electron migration from MnO_2 to Fe_3O_4 and are consistent with the conclusions from XPS.

Additionally, the CO-TPD of Fe_3O_4-0.3Mn is shown in Figure S6. There is one distinct CO desorption peak located at 100 °C for the Fe_3O_4-0.3Mn catalyst, which corresponds to the molecular CO adsorbed state, indicating weak CO adsorption.[55] Another weak CO desorption peak located in the 350−700 °C range is also observed for Fe_3O_4-0.3Mn, which can be ascribed to desorption of the dissociative CO, implying strong CO adsorption.[16,56] However, the intensity of the dissociative CO desorption peak for Fe_3O_4-0.3Mn is significantly weaker than that for Fe_3O_4@0.3MnO_2. These results further prove that the core−shell configuration enhances the ability of MnO_2 donating electrons to Fe_3O_4, which can reduce the electron vacancies in surface iron species and promote the dissociation and adsorption of CO, increasing chain growth.

3.1.6. Phase Transitions during Reaction. Mössbauer spectroscopy is widely considered as an effective characterization technique to distinguish and quantify complicated Fe species in Fe-based catalysts. The Mössbauer spectroscopy of the catalysts with different MnO_2 contents after the reaction can be deconvoluted into several sextets and a central doublet, which are shown in Figure S7. In addition, the spectral parameters are listed in Table 3. The sextets with hyperfine fields (Hhfs) of 480.49−485.65 and 435.97−450.49 kOe can be attributed to Fe_3O_4. The former represents the Fe^{3+} ions at the tetrahedral site (A site), while the latter represents the Fe^{3+} and Fe^{2+} ions at the octahedral site (B site) in Fe_3O_4. Furthermore, a superparamagnetic (spm) doublet with an IS value of approximately 0.17−0.66 mm/s can be assigned to the octahedral Fe^{3+} ion of the superparamagnetic state in small crystallites.[57] The fitted Hhf values of the 214.51−220.40, 174.57−188.51, and 97.98−109.98 kOe ranges are attributed to χ-Fe_5C_2.[58] In addition, the Hhf value of 187.86−195.75 kOe can be attributed to θ-Fe_3C.[59] Table 3 indicates that after the reaction, the samples comprise χ-Fe_5C_2, Fe_3O_4, and Fe^{3+} as the MnO_2 content is below 0.3. When the MnO_2 content increases from 0 to 0.2, the ratio of χ-Fe_5C_2 in the Fe species increases from 69.2 to 93.4%, while the ratio of Fe_3O_4 decreases from 24.3 to 4.2%. This result implies that Mn promotes the transformation of Fe_3O_4 to χ-Fe_5C_2 in the HTFT synthesis, which is consistent with the results of XRD. Notably, as the content of Mn increases to 0.3, the θ-Fe_3C phase emerges, while the Fe_3O_4 phase disappears. The Mössbauer spectroscopy of Fe_3O_4@0.2MnO_2 after 48 h of reaction in syngas does not present the θ-Fe_3C phase, which contradicts the XRD pattern of Fe_3O_4@0.2MnO_2 after the reaction that shows two weak θ-Fe_3C diffraction peaks. This contradiction can occur because it is difficult to fit the sextet of θ-Fe_3C in the spectrum because of the low θ-Fe_3C content when the content of Mn is 0.2. The content of θ-Fe_3C increases from 5.4 to 11.7% as the content of Mn increases from 0.3 to 0.4. In the meantime,

Table 3. Mössbauer Parameters of the Spent Samples[a]

		Mössbauer parameters			
catalyst	phase	IS (mm/s)	QS (mm/s)	Hhf (kOe)	area[b] (%)
Fe_3O_4	Fe_5C_2	0.24	−0.07	217.40	26.9
		0.18	−0.01	181.37	28.9
		0.27	0.15	102.46	13.4
	Fe_3O_4(A)	0.35	−0.04	485.65	14.6
	Fe_3O_4(B)	0.59	−0.46	450.49	9.7
	Fe^{3+}(spm)	0.36	1.04		6.5
Fe_3O_4@ 0.1MnO_2	Fe_5C_2	0.26	−0.12	215.49	27.6
		0.21	−0.01	182.02	34.1
		0.25	0.01	109.98	17.6
	Fe_3O_4(A)	0.38	−0.06	480.88	5.7
	Fe_3O_4(B)	0.59	0.08	435.97	5.8
	Fe^{3+}(spm)	0.34	1.00		9.2
Fe_3O_4@ 0.2MnO_2	Fe_5C_2	0.24	−0.07	214.51	38.5
		0.24	0.06	188.51	37.6
		0.12	0.05	104.46	17.3
	Fe_3O_4(A)	0.66	0.47	480.49	4.2
	Fe^{3+}(spm)	0.17	1.00		2.4
Fe_3O_4@ 0.3MnO_2	Fe_5C_2	0.26	−0.06	220.37	31.6
		0.20	−0.03	177.27	31.4
		0.20	−0.11	97.98	29.2
	Fe^{3+}(spm)	0.39	1.09		2.4
	Fe_3C	0.19	0.05	195.75	5.4
Fe_3O_4@ 0.4MnO_2	Fe_5C_2	0.26	−0.04	220.40	34.9
		0.17	−0.08	174.57	21.2
		0.20	−0.13	98.51	29.2
	Fe^{3+}(spm)	0.40	1.02		3.0
	Fe_3C	0.20	0.05	187.86	11.7

[a]Reaction condition: 340 °C, H_2/CO = 2, 1.5 MPa, 11 000 mL/(h·g_{Cat}), 48 h. [b]Measured at room temperature and max error = ±1%.

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Table 4. Activity and Product Distributions of the Samples[a]

catalyst	X_{CO} (%)	S_{CO_2} (%)	hydrocarbon distribution (%) CH_4	$C_2^=-C_4^=$	$C_2^0-C_4^0$	C_5^+	O/P	C balance (%)
Fe_3O_4	32.5	35.8	42.6	15.4	34.1	7.9	0.45	97.90
Fe_3O_4@0.1MnO_2	59.6	36.1	27.6	19.2	39.9	13.3	0.48	96.87
Fe_3O_4@0.2MnO_2	97.6	34.2	25.3	25.8	19.9	29.0	1.30	97.62
Fe_3O_4@0.3MnO_2	91.8	37.9	12.0	37.4	4.90	45.7	7.56	96.27
Fe_3O_4@0.4MnO_2	85.7	35.8	13.5	29.6	5.20	51.7	5.71	96.36
Fe_3O_4-0.3Mn	89.6	38.1	26.3	19.6	28.3	25.8	0.70	96.06

[a]Reaction condition: 340 °C, H_2/CO = 2, 1.5 MPa, 11 000 mL/(h·g_{Cat}), 48 h.

there is a continuous decrease in the amount of χ-Fe_5C_2 with the increase in the θ-Fe_3C amount. The θ-Fe_3C can transform into χ-Fe_5C_2 during the FT synthesis reaction.[60] However, the Mn promoter can affect the electronic state of surface carbonaceous species and lead to the formation of θ-Fe_3C.[59] In addition, we speculate that the confinement effect of the shell can restrain the conversion of θ-Fe_3C to χ-Fe_5C_2. The increase of the thickness of the shell arising from the increase in the content of Mn raises the θ-Fe_3C amount in the spent catalyst.

3.2. HTFT Reaction Performance. The prepared Fe_3O_4@MnO_2 core−shell nanocomposites and Fe_3O_4-0.3Mn catalyst were applied to the HTFT synthesis of light olefins. Table 4 sums up the HTFT activities and product distributions over the samples under industrially relevant conditions. The carbon balances are all higher than 96%, which indicates the accuracy of the experimental data. The CO conversion significantly increases from 32.5 to 97.6% as the content of Mn increases from 0 to 0.2. The characterization results of XRD, H_2-TPR, EDS-lining, and EDS-mapping mentioned above prove that Mn ions do not incorporate into the Fe lattice to form a solid solution. This result is contrary to the conclusions of other research stating that Mn could hinder the reduction and carbonization procedures of Fe-based catalysts and decrease the catalytic activity because of the enhanced interaction between Fe and Mn forming a solid solution.[27,61,62] Instead, the enhanced ability of MnO_2 donating electrons to Fe_3O_4 promotes the reduction and carburization of Fe oxides with the increase in the content of Mn, which is demonstrated by XPS, H_2-TPR, and CO-TPD. In addition, Mn reduces the grain size and promotes the dispersity of Fe_3O_4, increasing the BET surface area and improving the FT activity. However, the CO conversion is slightly reduced when the molar ratio of Mn is over 0.2. The reduction of CO conversion could be resulted from the diffusion limitation caused by the thicker MnO_2 layer, which can hinder the contact between reactants and Fe species. The selectivity toward CH_4 and $C_2^0-C_4^0$ is simultaneously decreased, while the selectivity for $C_2^=-C_4^=$ is significantly increased as the content of Mn increases from 0 to 0.3. Meanwhile, the value of O/P increases from 0.45 to 7.56, indicating the suppression of the hydrogenation reaction. It was reported that dissociated H from the surface of Fe species could spill to reducible MnOx through the pore channel of MnO_2, inhibiting the hydrogenation of the catalysts.[63] Furthermore, we speculate that the smaller pore size and the thicker MnO_2 shell may inhibit the rapid diffusion of the product and induce a higher probability of reabsorption, enhancing the C−C coupling ability. However, the selectivity to CH_4 and $C_2^0-C_4^0$ is slightly increased, while the selectivity for $C_2^=-C_4^=$ is significantly reduced when the content of Mn increases to 0.4. This result is attributed to the excess thickness of the MnO_2 layer inhibiting the contact between CO and Fe species. The variety of CO_2 selectivity over different Fe_3O_4@MnO_2 catalysts is not obvious. Ordinarily, Fe_3O_4 is identified as the active ingredient in the water−gas shift reaction. However, Liu et al.[64] proposed a new conclusion through the combination of experiments and a theoretical study. The authors verified that Fe_3O_4 was beneficial to the reverse water−gas shift reaction, resulting in a reduction of CO_2 selectivity. In contrast, the Boudouard mechanism occurring in the active χ-Fe_5C_2 phase plays a major role in the production of CO_2. By combining the results of Mössbauer spectroscopy, it can be found that there is no Fe_3O_4 phase in the spent Fe_3O_4@0.3MnO_2 and Fe_3O_4@0.4MnO_2 catalysts. However, there is the formation of CO_2 in the products, and the selectivity of CO_2 does not decrease significantly compared with that of other samples. We speculate that not only Fe_3O_4 but also χ-Fe_5C_2 facilitates the formation of CO_2 in our catalyst system. Compared with the selectivity of CO_2 of Fe_3O_4@0.3MnO_2, the CO_2 selectivity of Fe_3O_4@0.4MnO_2 slightly decreased from 37.9 to 35.8%, which is in pace with the reduction of the χ-Fe_5C_2 content from 92.2 to 85.3%. This result indicates that χ-Fe_5C_2 facilitates the formation of CO_2. The Fe_3O_4@0.3MnO_2 exhibits the highest light olefin selectivity of 37.4% at a CO conversion of 91.8%. This result is consistent with those of CO-TPD and H_2-TPD, indicating that Mn promotes the adsorption and dissociation of CO while suppressing the hydrogenation of olefins. However, when the content of Mn increases to 0.4, the hydrocarbon distribution shifts toward heavy hydrocarbons, while the $C_2^=-C_4^=$ selectivity decreases to 29.6%. We speculate that an increasing effect on the diffusion of products with the increase of the thickness of the MnO_2 layer plays a predominant role. A thicker shell leads to a higher possibility of reabsorption and carbon chain growth. The BET result indicates that the pore size exhibits a decreasing tendency with the increase in the content of Mn. The smaller pore size enhances the diffusion resistance of products and intermediates, which will cause product reabsorption. Therefore, these reabsorbed products will be induced to generate into longer-chain hydrocarbons due to the extended residence time in the catalysts with smaller pore sizes. By combining the above results of Mössbauer spectroscopy, it can be found that the amount of iron carbides continues to increase with the content of Mn, which is widely considered the active ingredient of the FT synthesis, indicating that Mn promotes the formation of iron carbides. In addition, Mn has promotional effects on the formation of θ-Fe_3C.[59] The amount of θ-Fe_3C increases from 5.4 to 11.7% as the content of Mn increases from 0.3 to 0.4. This result is consistent with the increase in C_5^+ selectivity. Therefore, we deduce that θ-Fe_3C

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has a positive correlation with C_5^+ selectivity, which is consistent with previous reports.[65]

In addition, the Fe_3O_4-0.3Mn sample synthesized by the coprecipitation process was applied to the HTFT reaction of $C_2^=-C_4^=$. Compared with the conversion of CO of Fe_3O_4@0.3MnO_2, the CO conversion of Fe_3O_4-0.3Mn is slightly decreased, which may be explained by the fact that the Fe-Mn spinel inhibits the reduction and carbonization of Fe_3O_4-0.3Mn. The selectivities of CO_2, CH_4, and light alkanes of the Fe_3O_4-0.3Mn are all higher than that of Fe_3O_4@0.3MnO_2. In contrast, the selectivity of $C_2^=-C_4^=$ and the value of O/P of the Fe_3O_4-0.3Mn are both inferior to that of Fe_3O_4@0.3MnO_2. These results are consistent with the conclusions from H_2-TPD and CO-TPD that the core−shell configuration enhances the ability of MnO_2 donating electrons to Fe_3O_4, which can reduce electron vacancies of iron species' surface and promote the dissociation and adsorption of CO, increasing chain growth. In contrast, the spinel structure does not have these characteristics.

The ASF model was employed to predict the hydrocarbon distribution of the products. Figure S9 depicts that the α value increases from 0.39 to 0.74 with the Mn content, indicating that Mn increases the chain growth ability. Noticeably, the α value of Fe_3O_4-0.3Mn is significantly lower than that of Fe_3O_4@0.3MnO_2. This result can be interpreted as indicating that the core−shell configuration enhances the ability of MnO_2 donating electrons to Fe_3O_4, which can reduce the electron vacancies of surface iron species and promote the dissociation and adsorption of CO while suppressing the hydrogenation reaction, thus increasing chain growth. For Fe_3O_4-0.3Mn prepared by the coprecipitation method, the strong interaction between Fe and Mn makes it difficult for electrons to transfer due to the presence of the Fe-Mn spinel phase. This result is consistent with the conclusions from XPS, H_2-TPD, and CO-TPD. A stability test was performed over the Fe_3O_4@0.3MnO_2 catalyst for 288 h. Figure 11 depicts that the CO conversion and product selectivity remain to stabilize, revealing that the catalyst has high catalytic performance and light olefin selectivity while maintaining excellent stability. The superior catalyst stability can be attributed to the confinement effect of the core−shell design.

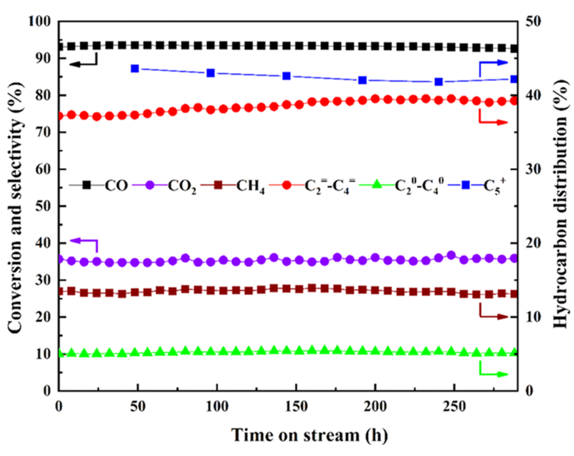

Figure 11. Stability test of the Fe_3O_4@0.3MnO_2 catalyst. Reaction conditions: 340 °C, H_2/CO = 2, 1.5 MPa, 11 000 mL/(h·g_{Cat}).

4. CONCLUSIONS

Overall, a one-pot hydrothermal process was adopted to synthesize Fe_3O_4@MnO_2 core−shell catalysts, which effectively convert syngas into $C_2^=-C_4^=$ with high activity and $C_2^=-C_4^=$ selectivity. The Fe_3O_4@0.3MnO_2 catalyst exhibits the highest $C_2^=-C_4^=$ selectivity of 37.4% at a CO conversion of 91.8%. Mn promotes the dispersity of Fe_3O_4 and reduces the grain size of Fe_3O_4, increasing the BET surface area of the sample and improving the FT activity. This core−shell structure enhances the ability of MnO_2 donating electrons to Fe_3O_4, promoting the reduction of the catalysts. CO-TPD and H_2-TPD indicate that Mn promotes the dissociation and adsorption of CO while suppressing the hydrogenation reaction. The smaller pore size and the thicker MnO_2 layer can extend the residence time of products and intermediates, promoting chain growth. In addition, the increase of the iron carbide amount during the HTFT reaction with the increase in the content of Mn enhances the catalytic activity.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acs.iecr.9b04221.

XRD patterns of Fe_3O_4-0.3Mn and Fe_3O_4@0.3MnO_2 after heat treatment; SEM images of as-synthesized samples; HRTEM images of spent catalysts; H_2-TPD profiles of H_2-reduced Fe_3O_4-0.3Mn and Fe_3O_4@0.3MnO_2 catalysts (PDF)

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Notes

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